



Province of the  
**EASTERN CAPE**  
EDUCATION

Iphondo leMpuma Kapa: Isebe leMfundo  
Provinsie van die Oos Kaap: Departement van Onderwys  
Porafensie Ya Kapa Botjahabela: Lefapha la Thuto

**NATIONAL  
SENIOR CERTIFICATE/  
NATIONALE  
SENIOR SERTIFIKAAT**

**GRADE/GRAAD 11**

**NOVEMBER 2025**

**TECHNICAL SCIENCES P2 (CHEMISTRY)/  
TEGNIJSE WETENSKAPPE V2 (CHEMIE)  
MARKING GUIDELINE/NASIENRIGLYN**

**MARKS/PUNTE: 75**

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This marking guideline consists of 7 pages./  
*Hierdie nasienriglyn bestaan uit 7 bladsye.*

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## QUESTION/VRAAG 1

- 1.1 D ✓✓ (2)
- 1.2 C ✓✓ (2)
- 1.3 A ✓✓ (2)
- 1.4 B ✓✓ (2)
- 1.5 C ✓✓ (2)
- [10]**

## QUESTION/VRAAG 2

- 2.1 2.1.1 A system which can exchange energy and matter with the surroundings. ✓✓  
*'n Sisteem wat deeltjies en energie met die omgewing kan uitruil.* (2)
- 2.1.2 The sum of the kinetic and potential energies of all the molecules of the system. ✓✓  
*Die som van die kinetiese en potensiële energie van al die molekules van die sisteem.* (2)
- 2.2 Temperature/Temperatuur ✓  
 Pressure/Druk ✓  
 Volume ✓ (3)
- 2.3 2.3.1 If heat energy is given to a system, it is used in increasing the internal energy of the system ✓ and doing work against external pressure. ✓  
*Indien warmte-energie aan 'n sisteem verskaf word, word dit gebruik om die interne energie van die sisteem te verhoog en om arbeid teen eksterne druk te verrig.* (2)
- 2.3.2  $\Delta Q = \Delta U + \Delta W$  ✓  
 $550\,000 = 145\,000 + \Delta W$  ✓  
 $\Delta W = 405\,000\text{ J OR/OF }405\text{ kJ}$  ✓ (3)
- 2.3.3 By the system ✓  
*Deur die sisteem*  
**APPLY NEGATIVE MARKING/PAS NEGATIEWE MERK TOE.**  
 The value of the work is positive ✓  
*Die waarde van die arbeid is positief* (2)
- [14]**

**QUESTION /VRAAG 3**

- 3.1 3.1.1 The amount heat required to increase the temperature of the whole substance by  $1^{\circ}\text{C}$  or  $1\text{K}$ . ✓✓  
*Die hoeveelheid warmte benodig om die temperatuur van die hele stof met  $1^{\circ}\text{C}$  of  $1\text{K}$  te laat styg.* (2)

3.1.2  $C = \text{cm}$  ✓  
 $= (400)(2)$  ✓  
 $= 800 \text{ J.K}^{-1}$  ✓ (3)

- 3.1.3 **POSITIVE MARK FROM QUESTION 3.1.2**  
**POSITIEWE NASIEN VANAF VRAAG 3.1.2**

**OPTION/OPSIE 1**

$$Q = \text{cm } \Delta T \quad \checkmark$$

$$= (400)(2)(40) \quad \checkmark$$

$$= 32\,000 \text{ J} \quad \checkmark$$

**OPTION/OPSIE 2**

$$C = \frac{Q}{\Delta T}$$

$$800 = \frac{Q}{40}$$

$$\therefore Q = 32\,000 \text{ J} \quad (3)$$

- 3.2 3.2.1 Thermometer/*Termometer* ✓ (1)

- 3.2.2 The amount of heat lost equals the amount of heat gained, when no heat is lost. ✓✓  
*Die hoeveelheid warmte wat afgegee word is gelyk aan die hoeveelheid warmte wat opgeneem word indien geen warmte verlore gaan nie.* (2)

3.2.3  $Q_{\text{gained/opgeneem}} = Q_{\text{lost/verlore}} \quad \checkmark$   
 $(\text{cm}\Delta T)_{\text{gained/opgeneem}} = (\text{cm}\Delta T)_{\text{lost/verlore}} \quad \left. \vphantom{\begin{matrix} Q_{\text{gained/opgeneem}} = Q_{\text{lost/verlore}} \\ (\text{cm}\Delta T)_{\text{gained/opgeneem}} = (\text{cm}\Delta T)_{\text{lost/verlore}} \end{matrix}} \right\} \text{ ANY ONE/ENIGE EEN}$   
 $4200 \checkmark \times 0,4 \checkmark \times 2 \checkmark = c \times 0,358 \checkmark \times 72 \checkmark$   
 $\therefore c = 130,35 \text{ J.kg}^{-1}.\text{K}^{-1} \sim 130 \text{ J.kg}^{-1}.\text{K}^{-1} \quad (6)$   
**[17]**

## QUESTION/VRAAG 4

- 4.1 Oxidation is the loss of electrons. ✓✓  
 Reduction is the gain of electrons. ✓✓  
*Oksidasie is die verlies van elektrone.*  
*Reduksie is die wins van elektrone.* (4)

- 4.2 4.2.1  $\underline{\text{Mg}}\text{Cl}_2$   
 $\text{Mg} + 2(-1) = 0$   
 $\therefore \text{Mg} = +2$  ✓✓

**NOTE/LET WEL:**

- Full marks awarded (  $\frac{2}{2}$  ) in Question 4.2 for answers only.

*Volpunte word toegeken (  $\frac{2}{2}$  ) in Vraag 4.2 vir slegs die antwoorde.* (2)

- 4.2.2  $\underline{\text{N}}\text{O}_3^-$   
 $\text{N} + 3(-2) = -1$   
 $\therefore \text{N} = +5$  ✓✓ (2)

- 4.2.3  $\text{Na}_2\underline{\text{C}}\text{O}_3$   
 $2(+1) + \text{C} + 3(-2) = 0$   
 $\therefore \text{C} = +4$  ✓✓ (2)

- 4.3 4.3.1 It involves the transfer of electrons (between substances). ✓✓  
*Dit behels die oordrag van elektrone (tussen stowwe).* (2)

- 4.3.2 Iron(III)ions/Yster(III)ione/ $\text{Fe}^{3+}$  ✓✓ (2)

- 4.3.3  $\underline{\text{Al}}$ /Aluminium ✓  
**APPLY NEGATIVE MARKING/PAS NEGATIEWE MERK TOE**  
 Oxidation number increases (from 0 to +3) ✓✓  
*Oksidasiegetal neem toe (vanaf 0 tot +3)* (3)

[17]

**QUESTION/VRAAG 5**

- 5.1 The decomposition of a substance when an electric current passes through it ✓✓  
*Die opbreek van 'n stof wanneer 'n elektriese stroom daardeur beweeg.*

**OR/OF**

The chemical process in which electrical energy is converted to chemical energy.  
*Die chemiese proses waarin elektriese energie in chemiese energie omgeskakel word.*

**OR/OF**

The use of electrical energy to produce a chemical change.  
*Die gebruik van elektriese energie om 'n chemiese verandering te bewerkstellig.* (2)

- 5.2 Copper(II)chloride/*Koper(II)chloried* ✓ (1)

- 5.3 Cell/ Battery/Power source ✓  
*Sel/ Battery/Kragbron* (1)

- 5.4 A substance that undergoes reduction. ✓✓  
*'n Stof wat self reduksie ondergaan.*

**OR/OF**

A substance that gains electrons.  
*'n Stof wat elektrone bygekry het.*

**OR/OF**

A substance that is reduced.  
*'n Stof wat self gereduseer word* (2)

- 5.5 Copper(II)ions/*Koper(II)ione*/ $\text{Cu}^{2+}$  ✓ (1)

5.6 Electrode/Elektrode Y ✓

**APPLY NEGATIVE MARKING/PAS NEGATIEWE MERK TOE**

It is the positive electrode. ✓✓

*Dit is die positiewe elektrode.*

**OR/OF**

In an electrolytic cell the anode is positive.

*In 'n elektrolitiese sel is die anode positief.*

**OR/OF**

Oxidation will take place at the anode/positive electrode/electrode Y.

*Oksidasie sal by die anode/positiewe elektrode/elektrode Y plaasvind.*

**OR/OF**

It is positively charged.

*Dit is positief gelaai.*

(3)

5.7 Cations/Katione ✓

(1)

5.8 Gas bubbles formed/bubbles formed/gas formed/chlorine gas formed ✓✓

*Gasborrels het gevorm/ borrels het gevorm/gas het gevorm/chloorgas het gevorm*

(2)

5.9 5.9.1  $\text{Cu}^{2+}_{(\text{aq})} + 2\text{e}^- \rightarrow \text{Cu}_{(\text{s})}$  ✓✓

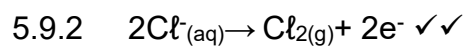
**MARKING CRITERIA/NASIENKRITERIA:**



**NOTE/LET WEL:**

- Do not penalise if the phases are omitted./  
*Moenie penaliseer indien die fases uitgelaat word nie.*

(2)



**MARKING CRITERIA/NASIEKRITERIA:**



**NOTE/LET WEL:**

- Do not penalise if the phases are omitted./  
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(2)  
[17]

**TOTAL/TOTAAL: 75**